

Set 5: MULTI-SENSOR ANOMALY & CONDITION MONITORING

Q1: EXCEL

Build a sensor monitoring workbook: per Plant compute hourly mean and std for Temperature/Vibration/Pressure, then flag anomaly rows where any sensor is outside mean ± 3 std.

Q2: EXCEL

Create a “Condition Index” using normalized (z-score) Temperature, Vibration, Pressure and weighted sum; rank MachineID by worst average Condition Index per Plant.

Q3: SQL

Return anomaly counts per Plant and per MachineID using threshold logic (mean ± 3 std) implemented via subqueries/CTEs.

Q4: SQL

Create view sensor_condition_2025 that aggregates per MachineID: AvgTemp, AvgVib, AvgPressure, AvgEnergyPerUnit, TotalDefects, MaintenanceCount, and ConditionBucket (Top 10% worst = Critical).

Q5: PYTHON

Isolation Forest (or z-score) anomaly detection using Temperature, Vibration, Pressure, EnergyConsumption, ProductionUnits. Output top 200 anomalies with timestamp and machine.

Q6: PYTHON

Build a composite “HealthScore” regression target you define from sensors and defects (document formula), then train a model to predict it and explain drivers.

Q7: POWER BI

Sensor monitoring dashboard: anomaly count KPI, trend of anomalies by day, machine ranking by Condition Index, slicers for Plant and MachineID.

Q8: POWER BI

Interactive condition exploration: synced line charts for Temperature/Vibration/Pressure over time with anomaly markers, plus table of anomaly events.